Brain MRI Diagnostic Report

VGG16 Transfer Learning · LIME Explainability · Automated Analysis

REPORT ID
BT-20260416-182836-F743

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Analysis summary

AI model:	VGG16 (transfer learning, 4-class head)
Input:	Single 2D slice, resized to 128×128 RGB
Pipeline:	Inference → occlusion XAI → LIME overlay
Classes:	Glioma · Meningioma · Pituitary · No Tumor

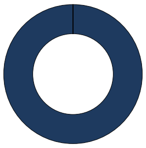
Diagnosis

PREDICTION

No Tumor Indicated

Raw label: No Tumor

Top-class confidence: 100.00%



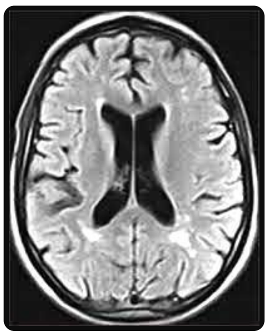
100.00%
CONFIDENCE

Class probabilities

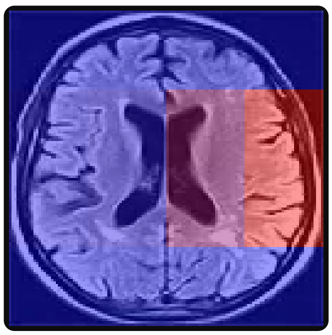
Meningioma	0.00%
Glioma	0.00%
Pituitary	0.00%
No Tumor	100.00%

Visualizations

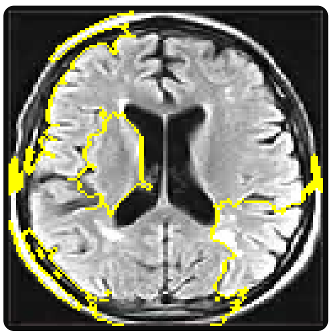
Original MRI



XAI heatmap (model attention)



LIME explanation



Findings & analysis

Model interpretation

The model did not assign high probability to any tumor class for this slice. This does not rule out pathology elsewhere in a full MRI study.

LIME explainability summary

LIME perturbs superpixels and fits a local linear model to approximate the classifier near this image. Boundaries mark regions that pushed the score toward the predicted class (No Tumor). Use alongside clinical context.

Report notes

- For research, education, or demonstration only.
- Not FDA-cleared; not a substitute for a radiologist.
- Single-slice uploads omit 3D context and sequence.
- Verify against the full MRI study and patient history.

Technical details

Architecture:	VGG16 backbone + dense classifier
Input resolution:	128 × 128 (preprocessed)
Explainability:	Occlusion sensitivity map; LIME
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Not a substitute for professional medical diagnosis

This AI-generated report is a preliminary analysis to assist clinicians and researchers. Final diagnosis and treatment decisions must be made by qualified healthcare professionals based on complete clinical evaluation.